Scaled Bohr donors, Fermi level & intrinsic crossover

Hydrogenic impurity derivation, valley-anisotropy honesty factor, Fermi shifts, and 560 K ceiling.

Hydrogenic impurity model (OL1, E4, P22 & P28)

An extra fifth valence electron of P orbiting a +e core inside crystal permittivity kappa = eps_r moves with band effective mass m*, rescaling Bohr's atom cleanly:

$$E_d = 13.6\ \text{eV}\,\frac{m^*/m_e}{\kappa^2}, \qquad a^* = a_0\,\frac{\kappa}{m^*/m_e} \quad (a_0 = 0.0529\ \text{nm})$$

$$\text{Si}\ (\kappa = 11.7,\ 0.26\,m_e):\ E_d = 26\ \text{meV},\ a^* = 2.4\ \text{nm}$$

$$\text{GaAs}\ (\kappa = 12.9,\ 0.067\,m_e):\ E_d = 5.5\ \text{meV},\ a^* = 10.2\ \text{nm}$$

WHY / Because a* = 2.4 nm spans ~4.4 cubic lattice cells (a_Si = 0.543 nm; ~9 bond lengths!), averaging over hundreds of Si atoms makes macroscopic dielectric screening kappa valid!

WATCH / Olympiad marking-rubric MUST-WRITE honesty line (Part 12): measured Si:P is 45 meV (1.7x higher) because silicon's 6 anisotropic conduction valleys & central-cell potential deviate from isotropic hydrogen!

Fermi-Dirac distribution & Fermi-level shift with doping

Occupation probability of state E is governed by Fermi-Dirac statistics; when E - EF >> kT, it reduces to Boltzmann factor exp(-(E - EF)/kT):

$$f(E) = \frac{1}{e^{(E-E_F)/(k_BT)}+1}, \qquad E_{F,i} = \frac{E_c+E_v}{2} + \frac{3}{4}k_BT\ln\left(\frac{m_h^*}{m_e^*}\right)$$

$$n = n_i\,e^{(E_F-E_i)/(k_BT)}, \qquad p = n_i\,e^{(E_i-E_F)/(k_BT)}$$

WHY / Multiplying n and p cancels EF identically to prove np = ni^2! n-doping shifts EF UP toward Ec by kT ln(Nd/ni); p-doping shifts EF DOWN toward Ev by kT ln(Na/ni)!

WATCH / At T -> 0 K in pure crystal, intrinsic Fermi level sits EXACTLY midway at (Ec + Ev)/2 regardless of mass asymmetry m_h* != m_e*!

Freeze-out (<120 K), extrinsic plateau & intrinsic ceiling

Plotting log n vs 1000/T across cryogenic to high temperature reveals three distinct physical regimes in doped silicon:

1. Freeze out $(k_BT \ll E_d):\ n \propto e^{-E_d/(2k_BT)}$
2. Extrinsic saturation (120 to 500 K): $n \approx N_d = \text{const!}$
3. Intrinsic $(n_i > N_d):\ n \approx n_i \propto T^{3/2}e^{-E_g/(2k_BT)}$

WHY / Notice factor 2 in BOTH slopes on an Arrhenius plot: steep high-T slope measures -Eg/(2k), shallow cryogenic slope measures -Ed/(2k)!

WATCH / In the extrinsic saturation window (near 300 K), n = Nd is constant while mu ~ T^(-3/2) falls — so lightly doped Si actually shows a SMALL POSITIVE dR/dT until ni wakes up!

When silicon devices forget their doping (OL12 exact)

Once thermal generation raises ni(T) up to dopant density Nd, p-n junctions lose rectification (V0 -> 0) and transistors fail completely:

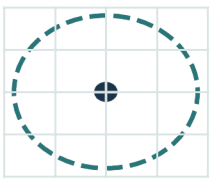
$$\left(\frac{T}{300}\right)^{3/2}\exp\left[\frac{E_g}{2k_B}\left(\frac{1}{300}-\frac{1}{T}\right)\right] = \frac{N_d}{n_i(300\ \text{K})} = 6.67\times10^4$$

$$\text{Si}\ (N_d = 10^{21}\ \text{m}^{-3}):\ T_{\max} \approx 560\ \text{K}\ (\approx 287^\circ\text{C})$$

WHY / Wide-bandgap semiconductors SiC (Eg = 3.26 eV) and GaN (Eg = 3.4 eV) push this thermal ceiling well past 600 deg C — why EV inverters & jet-engine sensors use SiC/GaN!

WATCH / Germanium (Eg = 0.66 eV) hits intrinsic washout right around ~90-100 deg C — why early 1950s Ge transistors failed in hot summer equipment and Si won!

Dielectric-screened effective-mass Bohr orbit of phosphorus in silicon



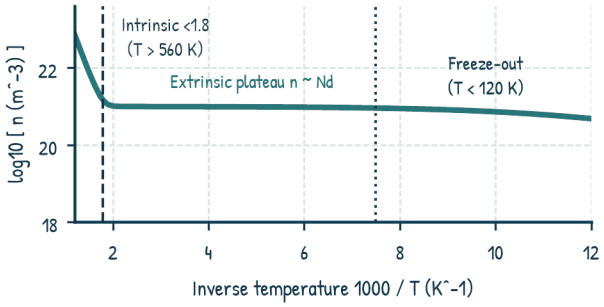
a* = 2.4 nm (~4.4 Si cells)

Scaled Hydrogenic Model

Ed = 13.6 (m*/m) / kappa^2
Si:P calc = 26 meV ~ kT
Measured = 45 meV (x1.7)

GaAs: Ed = 5.5 meV,
a* = 10.2 nm (shallow!)

Three temperature regimes of doped silicon: freeze-out, extrinsic, intrinsic



REMEMBER / Scaled Bohr gives Ed = 13.6 (m*/m) / kappa^2 = 26 meV ~ kT and a* = 2.4 nm in Si (measured 45 meV due to 6-valley anisotropy); Si loses doping above ~560 K!

Boltzmann diode derivation, Poisson step junction & varactors

Minority injection law, 2-sided vs 1-sided depletion audit, and voltage-tuned RF varactors.

Shockley equation from Boltzmann boundary law (OL2)

In equilibrium, minority electrons at the p-side depletion edge equal n_{p0} = n_{n0} exp(-eV0/kT). Forward bias V lowers barrier to V0 - V, boosting edge density exponentially:

$$n_p(-x_p) = n_{p0}\, e^{V/V_T} \Rightarrow \Delta n_p = n_{p0}(e^{V/V_T} - 1)$$

$$I = I_0(e^{V/V_T} - 1), \qquad I_0 = eA\left(\frac{D_p\, p_{n0}}{L_p} + \frac{D_n\, n_{p0}}{L_n}\right)$$

WHY / Using p_{n0} = ni^2 / Nd and n_{p0} = ni^2 / Na immediately proves WHY reverse leakage I0 scales as ni^2 and why heavily doping one side suppresses its reverse injection!

WATCH / Diffusion length L_n = sqrt(Dn tau_n) is how far injected minority electrons travel before recombining; a diode needs base thicker than L_n, whereas a BJT needs W_B << L_n!

Exact step-junction Poisson audit: 2-sided vs 1-sided (OL3/P29)

Audit distinction (OL3 & P29): total depletion width W = xp + xn across acceptor Na and donor Nd sides depends on REDUCED doping density N_eff:

$$W = \sqrt{\frac{2\varepsilon(V_0 + V_R)}{e\, N_{\text{eff}}}}, \qquad N_{\text{eff}} = \frac{N_a N_d}{N_a + N_d}$$

$$W_{1-\text{side}} = \sqrt{\frac{2\varepsilon(V_0 + V_R)}{eN}}, \qquad W_{\text{sym}} = \sqrt{2}\, W_{1-\text{side}}$$

WHY / In P29 (eps = 11.7 eps0, N = 10^21 m^-3, V0 + VR = 5.0 V, A = 1 mm^2): p+n one-sided gives W = 2.55 um, Cj = 40.7 pF; symmetric Na = Nd = N gives W = 3.60 um, Cj = 28.8 pF!

WATCH / Always state explicitly in Olympiad solutions whether you model a one-sided p^+n junction (N_eff = N) or symmetric junction (N_eff = N/2) — instant full rigor credit!

Varactor (variable-reactance) tuning diodes & 1/C^2 law

Small AC signal wiggle dV around reverse bias VR charges/discharges depletion edges separated by W(VR) exactly like a voltage-controlled parallel-plate capacitor:

$$C_j(V_R) = \frac{dQ}{dV_R} = \frac{\varepsilon A}{W(V_R)} = \frac{C_{j0}}{\sqrt{1 + V_R/V_0}}$$

$$\frac{1}{C_j^2} = \frac{2(V_0 + V_R)}{e\,\varepsilon\, A^2 N_{\text{eff}}} \quad (\text{STRICT STRAIGHT LINE vs } V_R!)$$

WHY / Plotting 1/Cj^2 against reverse bias VR in the lab measures BOTH built-in potential V0 (horizontal intercept -V0!) AND substrate doping N_eff (from slope)!

WATCH / Putting a varactor inside an LC tank f0 = 1 / (2 pi sqrt(L Cj)) gives f0 ~ (V0 + VR)^(+1/4) — how every PLL, FM modulator, and radio tuner tunes frequency electronically!

Linearly graded junctions & forward diffusion capacitance

Not every junction is a step function! Diffused junctions with rho(x) = e a x obey a cube-root law, while forward-biased diodes add a huge storage capacitance C_diff:

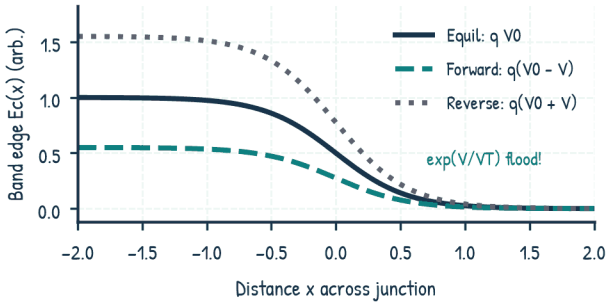
$$\text{Graded } \rho = eax: \quad W \propto (V_0 + V_R)^{1/3} \Rightarrow C_j \propto (V_0 + V_R)^{-1/3}$$

$$\text{Forward storage}: \quad Q_{\min} = \tau_T I \Rightarrow C_{\text{diff}} = \frac{dQ_{\min}}{dV} = \frac{\tau_T I}{V_T}$$

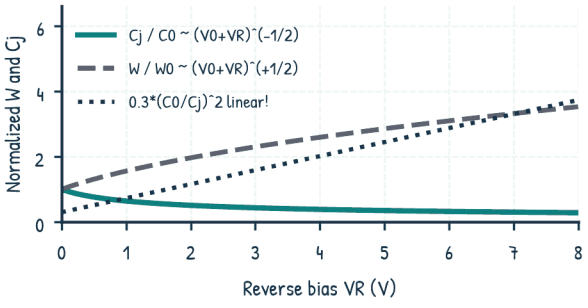
WHY / Stored minority charge Q_min = tau_T I takes transit/lifetime time tau_T to clear on sudden turn-off — why ordinary 1N4007 diodes fail at 100 kHz and Schottky diodes (majority carriers, tau_T ~ 0!) rule SMPS!

WATCH / Hyperabrupt varactors engineer doping N(x) ~ x^(-3/2) so Cj ~ (V0 + VR)^(-2), making LC resonant frequency f0 ~ 1/sqrt(Cj) STRICTLY LINEAR in control voltage VR!

Bias modulation of conduction-band barrier hill q(V0 - V) across a junction



Depletion width W(VR), varactor Cj(VR), and linear 1/Cj^2 extraction



REMEMBER / Watch 1-sided vs symmetric step junctions: W_1side = sqrt(2 eps V_tot / eN), whereas symmetric Na = Nd = N has W_sym = sqrt(4 eps V_tot / eN) = sqrt(2) W_1side!

Hall effect metrology, diode thermometry & 1.22 V reference

Lorentz carrier-sign determination, two-carrier Hall zero, -2 mV/K sensor, and Brokaw bandgap circuit.

Hall effect derivation, polarity & carrier extraction (OL6)

Pass current I_x along +x through rectangular slab (thickness t along B_z). Magnetic force q v_d x B pushes BOTH drift electrons (v_x < 0, q = -e) and drift holes (v_x > 0, q = +e) toward the SAME face!

$$|E_y| = v_d B_z = \frac{J_x B_z}{ne} \Rightarrow V_H = \frac{IB}{net}, \qquad R_H \equiv \frac{E_y}{J_x B_z} = \mp\frac{1}{ne}$$

WHY / OL6 & P33 check: I = 10 mA, B = 0.5 T, t = 0.5 mm, V_H = 62 mV => n = (0.01 x 0.5)/(1.6e-19 x 0.062 x 5e-4) = 1.01 x 10^21 m^(-3); doubling V_H halves n (P20: 31 mV -> 2.0 x 10^21 m^-3)!

WATCH / Notice slab WIDTH w cancels out of V_H = I B / (n e t)! Thin epitaxial films (tiny t) generate huge, easily measured millivolt Hall signals!

Combining Hall + conductivity & two-carrier Hall zero

Conductivity alone gives ONLY the product n mu; pairing a 4-probe resistivity test with a Hall measurement extracts carrier density n AND mobility mu SEPARATELY:

$$|R_H| = \frac{1}{ne}, \qquad \mu_H = |R_H|\,\sigma = \frac{V_H L}{W B V_{\text{long}}}$$

$$R_{H,\,\text{ambipolar}} = \frac{p\mu_h^2 - n\mu_e^2}{e(p\mu_h + n\mu_e)^2} \quad (\text{vanishes when } p\mu_h^2 = n\mu_e^2!)$$

WHY / Look at mu^2 in the numerator of R_H! Because mu_e = 2.81 mu_h in Si, intrinsic Si (n = p = ni) actually shows an N-TYPE negative Hall coefficient!

WATCH / Classic IPhO trap: in lightly p-doped silicon where p = (mu_e/mu_h)^2 n ~ 7.9 n, Hall voltage passes right through ZERO even though mobile carriers are abundant!

Diode & BJT thermometer: dV_BE/dT = -2 mV/K (OL5)

Driving a diode-connected BJT with constant current source I_C = const and differentiating V_BE(T) = V_T ln(I_C / I_0(T)) with I_0 ~ T^3 exp(-Eg / kT):

$$\frac{dV_{BE}}{dT} \approx \frac{V_{BE} - (E_g/e + 3V_T)}{T} \approx \frac{0.70 - 1.20}{300\ \text{K}} \approx -1.7 \text{ to } -2.1\ \text{mV/K}$$

WHY / Even though V_T = kT/e tries to raise V_BE, saturation current I_0(T) explodes exponentially across Eg, pulling V_BE DOWN linearly across -55 to +150 deg C (CTAT = Complementary To Absolute T)!

WATCH / Every CPU/GPU die embeds on-chip BJT thermal diodes reading this exact -2 mV/K slope to throttle clock frequency before silicon overheats!

Bandgap voltage reference: cancelling CTAT + PTAT = 1.22 V

Run two matched BJTs at fixed current-density ratio J1 / J2 = M (e.g. M = 10): their V_BE difference eliminates I_0 completely to create a pure +T signal (PTAT):

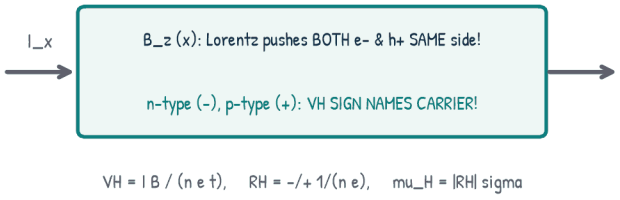
$$\Delta V_{BE} = V_{BE1} - V_{BE2} = V_T \ln M = \frac{k_B \ln M}{e} T \quad (\text{pure PTAT!})$$

$$V_{\text{ref}} = V_{BE} + K\Delta V_{BE} \Rightarrow \frac{dV_{\text{ref}}}{dT} = 0 \quad (1.22\ \text{V})$$

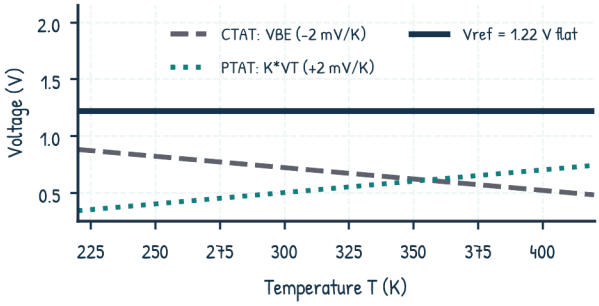
WHY / Summing falling V_BE (-2 mV/K, extrapolating to Eg/e at 0 K) + amplified rising K*dV_BE (+2 mV/K) locks V_ref right onto silicon's extrapolated 0 K bandgap ~1.22-1.25 V (P34)!

WATCH / Every precision IC regulator (LM317's famous 1.25 V reference, ADC refs, phone PMICs) rests directly on this CTAT + PTAT cancellation!

Hall effect slab geometry: Lorentz deflection names carrier sign and density



Cancelling CTAT (-2 mV/K) and PTAT (+2 mV/K) slopes in a 1.22 V reference



REMEMBER / Lorentz force deflects e- and h+ to the EXACT SAME transverse face — so Hall voltage polarity unambiguously distinguishes n-type from p-type!

Shockley-Queisser PV bound, shaded strings & tandem cells

Spectral loss trade-off (~33.7% at 1.34 eV), tandem solar stacks, hotspot physics, and bypass diodes.

Two fundamental spectral losses in single-junction PV (OL4)

Sunlight spans ~0.4 to 4.0 eV (~5800 K blackbody). A single semiconductor bandgap Eg extracts EXACTLY Eg (minus recombination loss) per absorbed photon:

$$h\nu < E_g: \text{ unabsorbed transmission loss } (0\%)$$

$$h\nu > E_g: \text{ excess } (h\nu - E_g) \text{ lost as phonons}$$

WHY / Tiny Eg absorbs everything but delivers tiny V_oc < Eg/e; huge Eg gives large V_oc but wastes nearly all solar photons -> product peaks sharply around Eg ~ 1.34 eV!

WATCH / In P30 Olympiad answers, you MUST name BOTH spectral losses PLUS radiative recombination (which forces V_oc < Eg/e and FF < 1) to derive the ~33.7% ceiling!

Why crystalline Si (1.12 eV) dominates & how tandems beat 33%

Silicon's 1.12 eV gap sits right beside the broad SQ plateau (~32% theoretical limit; ~27% lab record), backed by abundant SiO2 sand and wafer purity.

$$\text{Tandem}: \quad E_{g,\,\text{top}} \approx 1.7 \text{ eV}, \quad E_{g,\,\text{bot}} = 1.12 \text{ eV}$$

$$\eta_{\max}: \ 33.7\% \ (1\text{J}) \rightarrow 45\% \ (2\text{J}) \rightarrow 68\% \ (\infty\text{J})$$

WHY / Top wide-gap subcell harvests blue/green photons at HIGH voltage (~1.2 V) while letting red/IR pass transparently down into bottom Si subcell — crushing thermalisation loss!

WATCH / Focusing sunlight with lenses/mirrors (concentrator PV, C suns) boosts I_L -> raises V_oc by V_T ln C, lifting theoretical infinite-junction limit toward 86.8%!

Shaded series solar cell hotspot crisis (OL10 & P35)

Solar modules wire N cells (e.g. 60 or 72) IN SERIES so voltages add to ~35-40 V. If ONE single leaf shades cell #k, KCL forces identical current through every cell:

$$I_{\text{str}} > I_{L,\,\text{sh}} \Rightarrow V_{\text{shaded}} = V_{\text{load}} - \sum_{j \neq k} V_j < 0$$

$$P_{\text{hotspot}} = |V_{\text{shaded}}|\, I_{\text{string}} \quad \text{(tens of watts in ONE cell!)}$$

WHY / All N - 1 sunlit cells gang up like a 30 V battery ramming 8 A BACKWARD into the one shaded cell — baking EVA encapsulant and cracking glass!

WATCH / Without protection, either open-circuit load drops string power to the worst cell OR low load resistance drives destructive reverse avalanche breakdown!

Antiparallel bypass diodes & stepped multi-peak I-V curves

Wiring a Schottky bypass diode in ANTI-PARALLEL across each sub-string of M cells (typically 20-24 cells per diode, 3 diodes per 60/72-cell panel) saves both power and hardware:

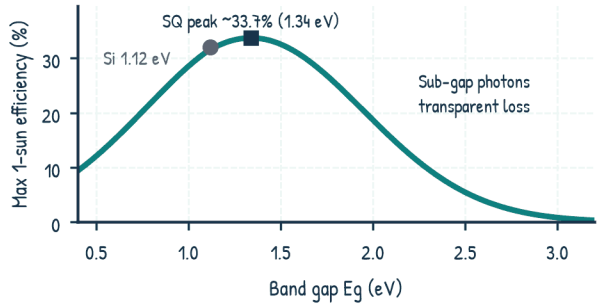
$$\text{Full sun } (V_{\text{blk}} > 0): \text{ bypass diode OFF}$$

$$\text{Shaded } (V_{\text{blk}} \approx -0.4 \text{ V}): \text{ bypass diode ON!}$$

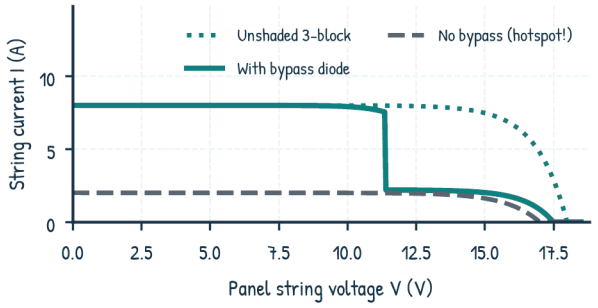
WHY / Full 8 A string current detours cleanly around the shaded sub-string: the module loses ONLY 1/3 of voltage (~65% power saved!) instead of collapsing by 80-90%!

WATCH / Signature on panel I(V) curve (P35c): K distinct current steps / local power peaks — which is why solar inverters run global MPPT sweeps!

Single-junction Shockley-Queisser limit (~33.7% near Eg = 1.34 eV)



Series PV string current collapse under shading vs bypass-diode step recovery



REMEMBER / Single-junction PV cells lose sub-gap red photons (unabsorbed) and excess above-gap blue energy (phonons), capping 1-sun efficiency near ~33.7% at Eg ~ 1.34 eV!

Shot noise, Johnson noise, NEP & LED load-line stability

Poisson discrete-charge floor, resistor thermal noise, minimum detectable power, and OL9 design.

Quantum shot noise from Poisson arrival statistics (OL8, P32)

Current I consists of discrete independent electrons e crossing a potential barrier at mean rate N / tau = I / e. Over observation window tau = 1 / (2 df), count variance is N:

$$\sigma_I^2 = (\tfrac{e}{\tau})^2 N = \tfrac{e^2}{\tau^2}(\tfrac{I\tau}{e}) = 2eI\,\Delta f \Rightarrow i_{\text{rms}} = \sqrt{2eI\,\Delta f}$$

WHY / OL8 & P32 checks: (1) I = 1 uA, df = 1 MHz => i_rms = 0.566 nA; (2) I = 4 uA, df = 2 MHz => i_rms = 1.60 nA! Narrowing receiver bandwidth df by 4x cuts rms noise in HALF!

WATCH / Dark current I_d and background light I_bg add uncorrelated shot noise even when signal modulates around zero: total shot variance is 2 e (I_sig + I_dark + I_bg) df!

Johnson-Nyquist resistor noise & photoreceiver SNR / NEP

Load resistor R_L at temperature T adds thermal Johnson current noise independent of DC current; total receiver noise sums mean-square powers:

$$i_{n,\,\text{tot}} = \sqrt{(2e(I_{\text{photo}} + I_{\text{dark}}) + \tfrac{4k_BT}{R_L})\Delta f}$$

$$\text{SNR} = \frac{I_{\text{photo}}^2}{i_{n,\,\text{tot}}^2}, \qquad \text{NEP} = \frac{i_{n,\,\text{tot}}|_{\Delta f = 1\ \text{Hz}}}{\mathcal{R}_\lambda} \quad (\text{W}/\sqrt{\text{Hz}})$$

WHY / Shot noise beats thermal noise once I*R_L > 2 k_B T / e = 51.7 mV! Any voltage drop > 52 mV across load resistor R_L puts the receiver in the quantum shot-noise regime!

WATCH / Never add rms currents linearly (i1 + i2)! Independent Gaussian/Poisson noise sources ALWAYS add IN QUADRATURE: sqrt(i_shot^2 + i_thermal^2)!

Why LEDs MUST have current-limiting ballast (OL9)

An LED is an exponential diode (I ~ exp(V / VT)), NOT an ohmic bulb! Connecting a 2.0 V LED directly across a stiff voltage rail invites instant thermal runaway:

$$R_{\text{series}} = \frac{V_{CC} - V_{\text{LED}}}{I_{\text{target}}} \quad (\text{OL9}: \ \frac{5.0 - 2.0\ \text{V}}{20\ \text{mA}} = 150\ \Omega, \ P_R = 60\ \text{mW})$$

WHY / Without R_series, a +100 mV supply bump or +50 K junction warm-up (-100 mV knee shift!) boosts LED current by 10^(100/60) ~ 46x — melting the bond wire in milliseconds!

WATCH / Never wire two bare LEDs directly IN PARALLEL off one shared resistor! The slightly warmer LED drops V_LED by 2 mV/K, hogs all current, overheats further, and cascades to destruction!

Small-signal sensitivity: voltage source vs series resistor

Compare fractional current sensitivity dI / I to a diode knee shift dV_LED under stiff voltage drive vs resistor-ballasted drive:

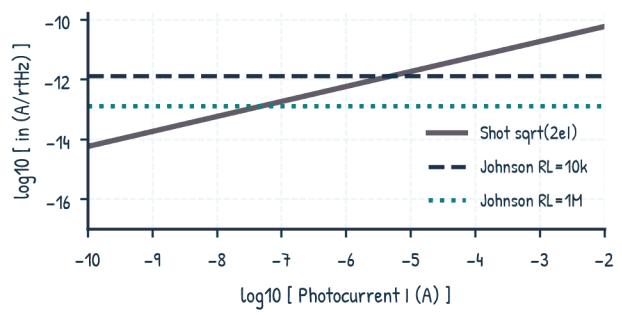
$$\text{Stiff V drive}: \ \frac{\delta I}{I} = -\frac{\delta V_{\text{LED}}}{V_T} \approx -39 \times \delta V_{\text{LED}} \ (\text{in volts!})$$

$$\text{With } R_s: \ \frac{\delta I}{I} = -\frac{\delta V_{\text{LED}}}{IR_s + V_T} \approx -\frac{\delta V_{\text{LED}}}{V_{CC} - V_{\text{LED}}} \ (115\text{x gentler!})$$

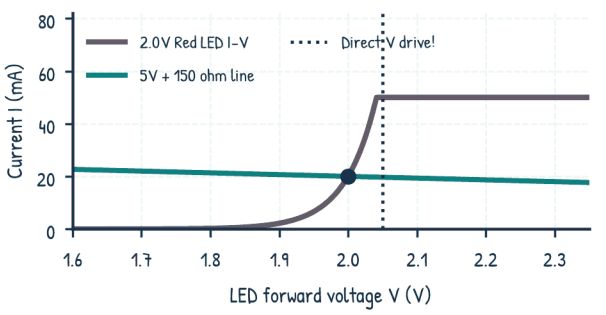
WHY / Dropping 3.0 V across R_s makes loop slope resistance I R_s + V_T = 3.026 V instead of 0.026 V — turning a wild exponential device into a rock-steady 3% current source!

WATCH / High-power white LED headlamps replace wasteful series resistors with buck switch-mode constant-current drivers (>92% efficiency vs 40% linear resistor)!

Quantum shot noise floor sqrt(2eI) vs Johnson-Nyquist resistor noise



Why LEDs need a series resistor: linear load line vs exponential runaway



REMEMBER / Shot noise i_rms = sqrt(2 e I df) comes from discrete charge quanta e; Johnson noise i_th = sqrt(4 kT df / R) comes from thermal agitation — add variances i_tot^2 = i_shot^2 + i_th^2!

Moore's quantum wall, oxide tunnelling & chip heat budget

Heisenberg confinement spread, WKB gate leakage, high-k FinFETs, and 500–800 W CMOS power.

Quantum wall #1: Heisenberg confinement energy (OL7, P31)

Squeezing channel/well dimension dx to a few nanometres gives carriers minimum zero-point kinetic energy spread via dx * dp >= hbar:

$$\Delta E \sim \frac{\hbar^2}{2m_e(\Delta x)^2}: \ 1.5 \text{ meV } (5 \text{ nm}), \ 4.2 \text{ meV } (3 \text{ nm})$$

$$\text{With } m^* = 0.19\, m_e: \ \Delta E(3 \text{ nm}) \approx 22 \text{ meV} \sim k_B T$$

WHY / Carrier de Broglie wavelength at 300 K is h / sqrt(3 m* kT) ~ 12 nm — smaller than a modern transistor pitch, so 2D/1D quantum subbands replace bulk classical drift!

WATCH / Quote BOTH free-electron mass result (4.2 meV at 3 nm, P31a exact!) AND note that light effective mass m* raises confinement right up to thermal kT = 26 meV!

Quantum wall #2: WKB gate-oxide tunnelling leakage

Scaling planar MOSFET capacitance C_ox = eps_ox A / d_ox once forced SiO2 thickness down to d_ox ~ 1.2 nm (~5 atomic layers!), where WKB tunnelling explodes:

$$T_{\text{WKB}} \sim \exp\left(-2\, d_{\text{ox}} \frac{\sqrt{2m_{\text{ox}}^* \Phi_B}}{\hbar}\right) \sim e^{-11.4\, d_{\text{ox}}(\text{nm})}$$

WHY / Every 0.2 nm thinning multiplies gate leakage current by ~10x! At 1.2 nm, 10^9 "OFF" transistors leak tens of amps straight through the gate insulator!

WATCH / How did semiconductor physics beat both walls (P31c)? (1) High-k HfO2 (kappa ~ 25 vs 3.9 -> 6x thicker physical wall for same C_ox!) + (2) 3D FinFET / Gate-All-Around nanosheets!

CMOS capacitor charging energetics & cycle audit (OL11, P36)

Reconciling single-transition vs full-cycle formulas cleanly from exact capacitor physics:

$$\text{Per edge } (0 \to 1 \text{ or } 1 \to 0): \ E_{\text{diss}} = \tfrac{1}{2} C V_{DD}^2$$

$$\text{Full cycle } (0 \to 1 \to 0): \ P_{\text{dyn}} = \alpha_{\text{sw}} N C V_{DD}^2 f$$

WHY / Audit clarity: if N gates each make 1 transition per clock period (activity alpha_sw = 0.5), P = (1/2) N C V^2 f = 500 W (OL11: N=10^9, f=1 GHz, C=1 fF, V=1 V) & 810 W (P36)!

WATCH / Making transistor channel R -> 0 does NOT reduce switching energy 1/2 CV^2 — it only burns the exact same 1/2 CV^2 faster in shorter RC time!

Why CPU clocks stalled near ~4–5 GHz (Dennard breakdown)

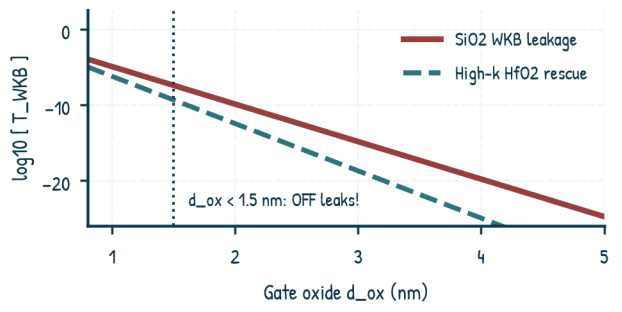
Total chip thermal dissipation combines dynamic switching power plus subthreshold leakage across billions of transistors:

$$P_{\text{total}} = \alpha_{\text{sw}} N C V_{DD}^2 f + N V_{DD} I_0\, e^{-V_{\text{th}}/(\eta V_T)}$$

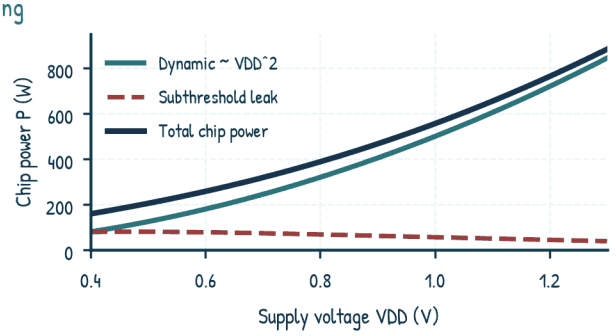
WHY / Because P_dyn scales quadratically with V_DD^2, dropping V_DD from 5 V -> 0.8 V cut switching energy by 39x! Below ~0.7 V, lowering threshold V_th makes exponential subthreshold leakage explode (60 mV/dec floor!) — freezing single-core GHz ~2005 and birthing multi-core CPUs!

WATCH / Notice how the ENTIRE chapter closes one circle: the 60 mV/decade Boltzmann limit of Sheet 05 IS the subthreshold leakage wall of 2026 2-nm chips!

WKB gate-oxide tunnelling probability and high-k HfO2 scaling rescue



Dynamic CV^2f vs subthreshold leakage trade-off across CMOS supply scaling



REMEMBER / Charging capacitor C through resistor to VDD draws CV^2 from battery: 1/2 CV^2 stored, 1/2 CV^2 burned on rise + 1/2 CV^2 burned on discharge = CV^2 per full 0->1->0 cycle!